Week- 4

```
(venv) PS C:\Users\ksrin\OneDrive\Desktop\KL\GEN AI\week 4> python tokenizer_comparison.py
Loading BPE - GPT-2...
Loading WordPiece - Multilingual BERT...
Loading SentencePiece - XLM-R...

======================================================================
LANGUAGE: English
INPUT TEXT: Artificial intelligence is changing the world.
======================================================================

BPE - GPT-2
Tokens     : ['Art', 'ificial', 'Ġintelligence', 'Ġis', 'Ġchanging', 'Ġthe', 'Ġworld', '.']
Token IDs  : [8001, 9542, 4430, 318, 5609, 262, 995, 13]
Token count: 8

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'Arti', '##ficial', 'intelligence', 'is', 'changing', 'the', 'world', '.', '[SEP]']
Token IDs  : [101, 90841, 69229, 30151, 10124, 43068, 10105, 11356, 119, 102]
Token count: 10

SentencePiece - XLM-R
Tokens     : ['<s>', '▁Artificial', '▁', 'intelligence', '▁is', '▁changing', '▁the', '▁world', '.', '</s>']
Token IDs  : [0, 231996, 6, 130687, 83, 151134, 70, 8999, 5, 2]
Token count: 10

======================================================================
LANGUAGE: Telugu
INPUT TEXT: కృత్రిమ మేధ ప్రపంచాన్ని మారుస్తోంది
======================================================================

BPE - GPT-2
Tokens     : ['à', '°', 'ķ', 'à', '±', 'Ĥ', 'à', '°', '¤', 'à', '±', 'į', 'à', '°', '°', 'à', '°', '¿', 'à', '°', '®', 'Ġ', 'à', '°', '®', 'à', '±', 'ĩ', 'à', '°', '§', 'à', '°', '.', 'à', '±', 'į', 'à', '°', '.', 'à', '±', 'ģ', 'Ġ', 'à', '°', 'ª', 'à', '±', 'į', 'à', '°', '°', 'à', '°', 'ª', 'à', '°', 'Ĥ', 'à', '°', 'Ļ', 'à', '°', '¾', 'à', '°', '¨', 'à', '±', 'į', 'à', '°', '¨', 'à', '°', '¿', 'Ġ', 'à', '°', '®', 'à', '°', '¾', 'à', '°', '°', 'à', '±', 'ģ', 'à', '°', '.', 'à', '±', 'į', 'à', '°', '¤', 'à', '±', 'ĩ', 'à', '°', 'Ĥ', 'à', '°', '¦', 'à', '°', '¿', '.']
Token IDs  : [156, 108, 243, 156, 109, 225, 156, 108, 97, 156, 109, 235, 156, 108, 108, 156, 108, 123, 156, 108, 106, 220, 156, 108, 106, 156, 109, 229, 156, 108, 100, 156, 108, 116, 156, 109, 235, 156, 108, 116, 156, 109, 223, 220, 156, 108, 103, 156, 109, 235, 156, 108, 108, 156, 108, 103, 156, 108, 224, 156, 108, 248, 156, 108, 122, 156, 108, 101, 156, 109, 235, 156, 108, 101, 156, 108, 123, 220, 156, 108, 106, 156, 108, 122, 156, 108, 108, 156, 109, 223, 156, 108, 116, 156, 109, 235, 156, 108, 97, 156, 109, 233, 156, 108, 224, 156, 108, 99, 156, 108, 123, 13]
Token count: 112
```

```
WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'క', '##ృ', '##త్ర', '##మ', 'బ', '##ుద్ధ', '##మ', '##త్తు', 'దు', '##న్య', 'బ', '##ద్ద', '##న', '##ోంది', '.', '[SEP]']
Token IDs  : [101, 1201, 111339, 88046, 24696, 1224, 17729, 99288, 15370, 39179, 94504, 23769, 1224, 27374, 15370, 68450, 119, 102]
Token count: 18

SentencePiece - XLM-R
Tokens     : ['<s>', '_', 'కృత్', 'రి', 'మ', '_బు', 'ద్ధ', 'మై', '_దు', 'న్య', '_బద్ద', 'ంగా', '_ఉం', 'ది', '.', '</s>']
Token IDs  : [0, 6, 151424, 91445, 3071, 22290, 11600, 129461, 86457, 6258, 169610, 104528, 5, 2]
Token count: 14

======================================================================
LANGUAGE: Hindi
INPUT TEXT: कृत्रिम बुद्धिमत्ता दुनिया को बदल रही है।
======================================================================

BPE - GPT-2
Tokens     : ['à¤', 'ķ', 'à¥', 'ĥ', 'à¤', 'ª', 'à¥', 'į', 'à¤', '°', 'à¤', '¿', 'à¤', '®', 'Ġà¤', '¬', 'à¥', 'ģ', 'à¤', '¦', 'à¥', 'į', 'à¤', '§', 'à¤', '¿', 'à¤', '®', 'à¤', 'ª', 'à¥', 'į', 'à¤', 'ª',
 'à¤¾', 'Ġà¤', '¦', 'à¥', 'ģ', 'à¤', 'Ļ', 'à¤', '¿', 'à¤', 'Ļ', 'à¤¾', 'Ġà¤', 'ķ', 'à¥', 'Ī', 'Ġà¤', '¬', 'à¤', '¦', 'à¤', '²', 'Ġà¤', '°', 'à¤', '¹', 'à¥', 'Ģ', 'Ġà¤', '¹', 'à¥', 'Ī', 'à¥', 'ª']
Token IDs  : [11976, 243, 24231, 225, 11976, 97, 24231, 235, 11976, 108, 11976, 123, 11976, 106, 28225, 105, 24231, 223, 11976, 99, 24231, 235, 11976, 100, 11976, 123, 11976, 106, 11976, 97, 24231, 235
, 11976, 97, 48077, 28225, 99, 24231, 223, 11976, 101, 11976, 123, 11976, 107, 48077, 28225, 243, 24231, 233, 28225, 105, 11976, 99, 11976, 110, 28225, 108, 11976, 117, 24231, 222, 28225, 117, 24231, 2
30, 24231, 97]
Token count: 68

WordPiece - Multilingual BERT
Tokens     : ['[CLS]', 'क', '##ृ', '##त्र', 'ब', '##ुद्ध', '##म', '##त्ता', 'दुन', 'को', 'ब', '##दल', 'रही', 'है', '।', '[SEP]']
Token IDs  : [101, 865, 37500, 18321, 50419, 887, 48010, 50419, 94034, 83874, 11267, 887, 99989, 57203, 10569, 920, 102]
Token count: 17

SentencePiece - XLM-R
Tokens     : ['<s>', '_कृत्', 'रि', 'म', '_बुद्ध', 'म', 'त्ता', '_दुन', '_को', '_बदल', '_रही', '_है', '।', '</s>']
Token IDs  : [0, 74707, 97885, 1920, 105202, 1920, 54297, 29405, 629, 14512, 8906, 460, 125, 2]
Token count: 14
(venv) PS C:\Users\ksrin\OneDrive\Desktop\KL\GEN AI\week 4>
```